



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- Summary of methodologies
 - Data Collection with API
 - Data Collection with Web Scrapping
 - Exploratory Data Analysis using SQL
 - Exploratory Data Analysis with Data Visualization
 - Building an interactive map with Folium
 - Building a Dashboard with Plotly
 - Predictive Analysis - Classification
- Summary of all results
 - EDA Results
 - Plotly dashboard results
 - Predictive Analysis results

Introduction

- Project background and context
 - For decades now we have been exploring the space and how to reach it, and SpaceX has opened a new paradigm: commercial space travel. Falcon 9 rocket launches have been remarkable for research due to its first stage reuse capabilities. Determination of the viability of the stage 1 can help defining the cost of a future launch. This data is available, and we can study the conditions on which these launchings are successful.
- Problems you want to find answers
 - How do conditions regarding payload mass, launch site, orbits, number of flights, booster version, etc., impact on the landing of a Falcon 9 launch.
 - Correlations between variables to understand the process and its possible liabilities
 - How has launching evolution shifted over time
 - Can Machine Learning help with predicting the outcome of a launching?

Section 1

Methodology

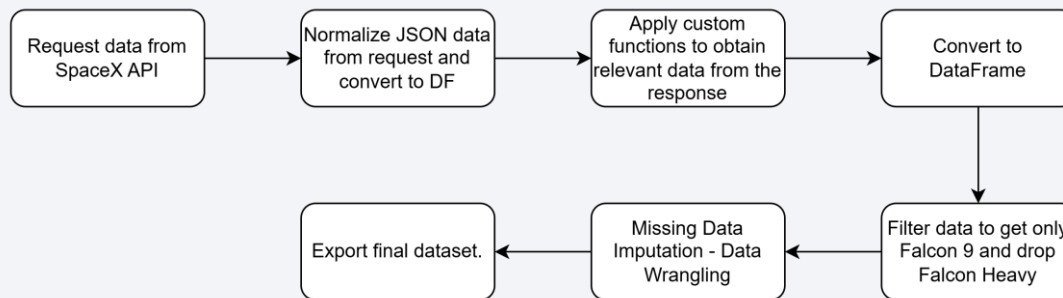
Methodology

Executive Summary

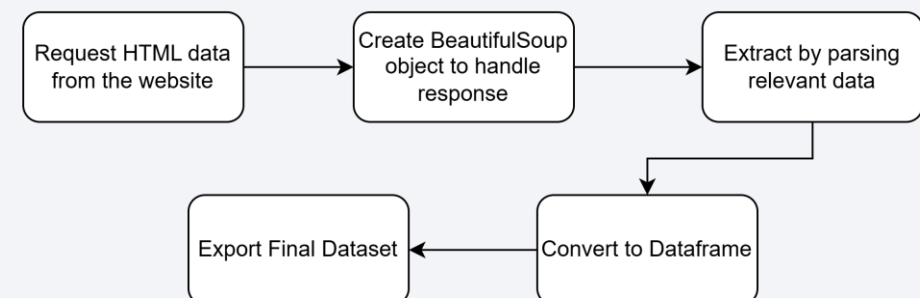
- Data collection methodology:
 - Data was collected using the SpaceX API and Web Scrapping from Wikipedia
- Perform data wrangling
 - Data was filtered to need, feature engineering was performed, missing values were treated by mean imputation, categorical variables treatment with encoding.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - The target is defined, then scalers are used for data normalization, segmentation of the dataset for train/test, training and tuning of LR, SVC, DT and KNN using grid search and cross validation. Then, scoring models by accuracy.

Data Collection

- The data for this project was obtained by consuming the SpaceX API and by web-scraping the Falcon 9 and Falcon Heavy Launches page in Wikipedia.



Process of SpaceX API Consumption -
[GitHub to Notebook](#)

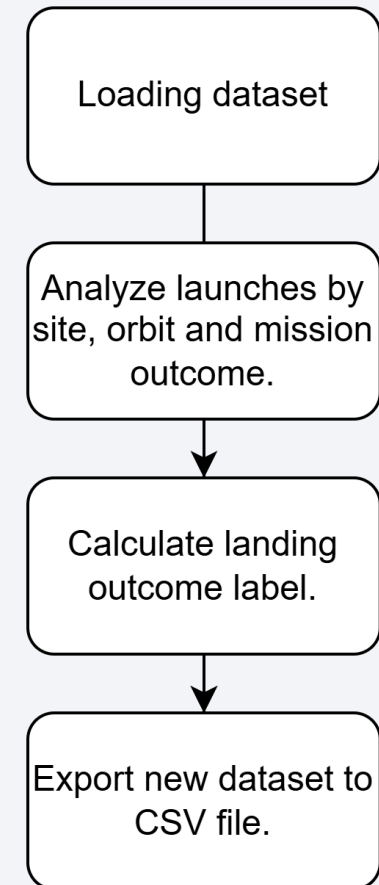


Process of Web Scrapping from Wikipedia
[GitHub to Notebook](#)

Data Wrangling

- The main target for wrangling was handling non successful booster landing. There were several use cases to know whether this had or hadn't happened. For instance,
 - True Ocean and False Ocean represent if the outcome was successfully landed to a specific region of the ocean.
 - True RTLS or False RTLS are about landings on groundpads.
 - True ASDS and False ASDS regard landings on drone ships.
- Considering this, a label is created to be set as 1 if landing successful, 0 if unsuccessful

[Link to GitHub Notebook](#)



EDA with Data Visualization

- In this EDA, we used:
 - Scatter plots: Flight Number vs. Launch Site, Payload vs Launch Site, Flight Number vs Orbit Type, Payload vs Orbit Type
 - Bar charts: Success Rate by Orbit Type
 - Line charts: Average Launch Success by year.
- Why?
 - Scatter plots help analyzing correlation between two variables, and you can use a hue to categorize.
 - Bar charts help understand frequency and compare discrete categories.
 - Line charts show trend changes
- [Link to Github Notebook](#)

EDA with SQL

- The SQL queries performed for this EDA stage were:
 - Display the name of unique launch sites
 - Display 5 records where launch sites begin with the string 'CCA'
 - Display the total payload mass carried by boosters launched by NASA (CRS)
 - Display average payload mass carried by booster version F9 v1.1
 - List the date when the first successful landing outcome in ground pad was achieved.
 - List the names of the booster which have success in drone ship and have payload mass greater than 4000 but less than 6000
 - List the total number of successful and failure mission outcomes
 - List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function
 - List the records which will display the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015
 - Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
- [Link to GitHub Notebook](#)

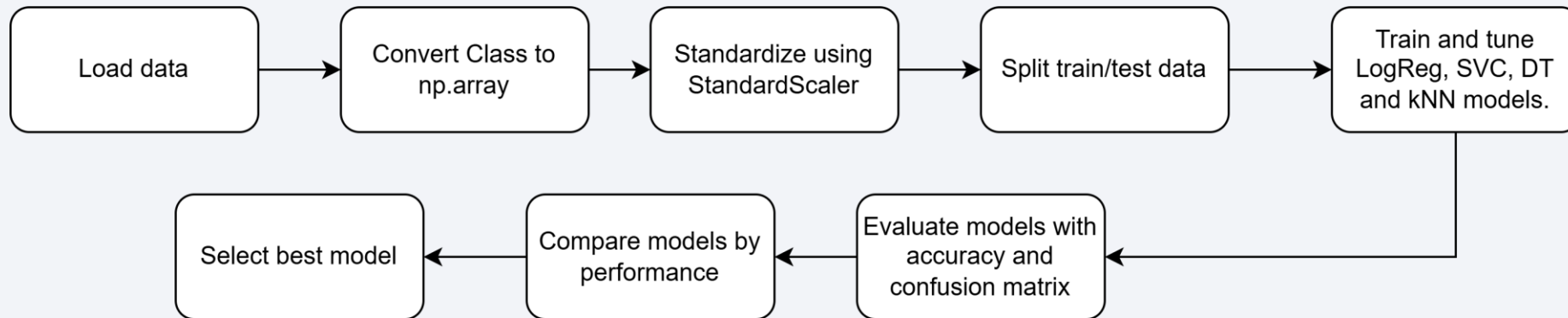
Build an Interactive Map with Folium

- Markers of All Launch Sites
- MarkerCluster() for success and failure launches
- Calculated distances between a launch site to nearby coastlines, railroads, highways, cities. With a plotly polyline()
- The map would be useful to estimate risks and liabilities when performing a launch.
- [Link to GitHub](#)

Build a Dashboard with Plotly Dash

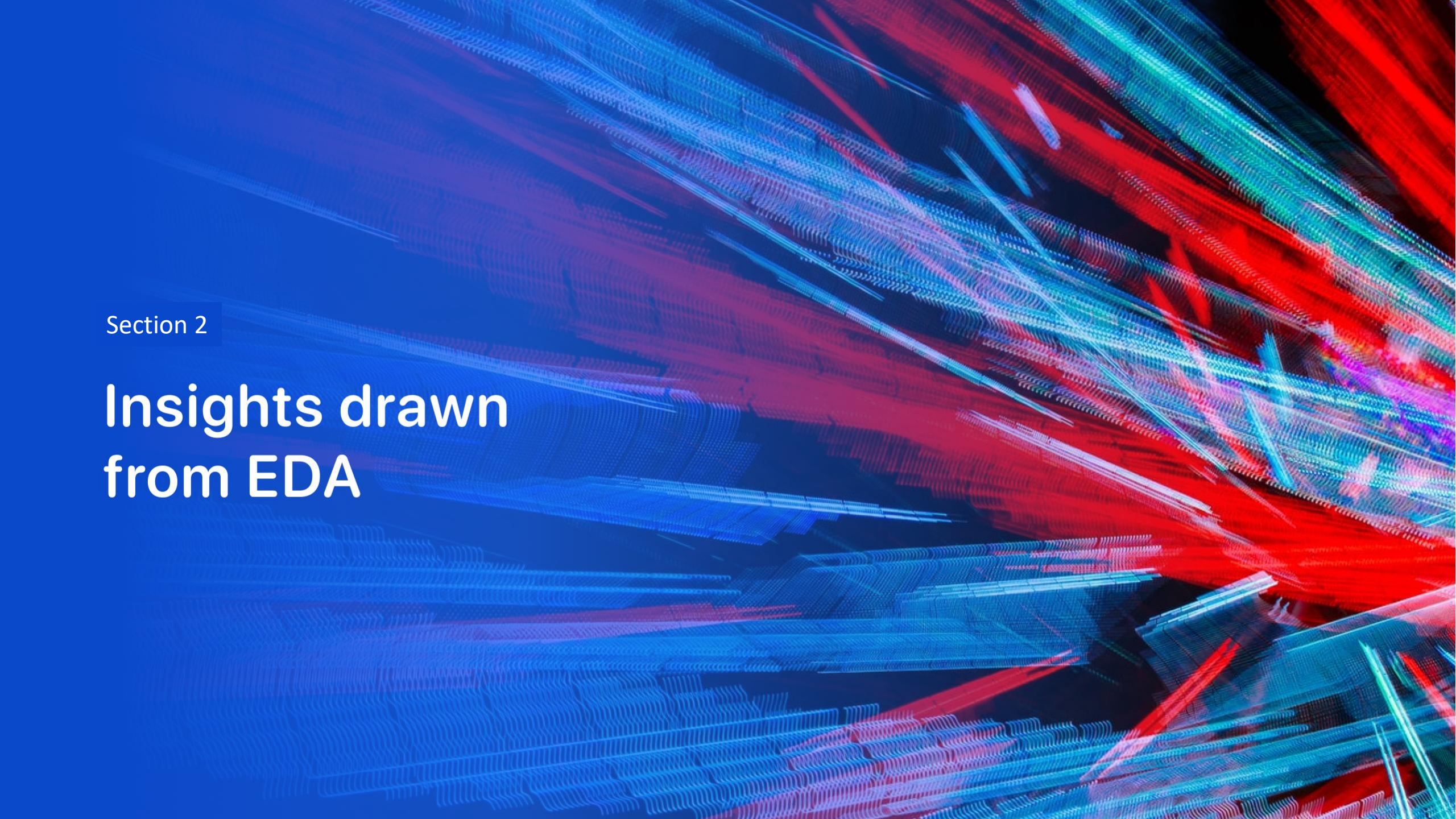
- A dropdown list was created including all sites.
- There's a pie chart available showing successful launches in every and any site.
- There's a slider to filter payload mass range and help identify visual patterns.
- There's a scatter chart to visualize how payload may be correlated with the outcome for the selected location.
- [Link to Github](#)

Predictive Analysis (Classification)



Results

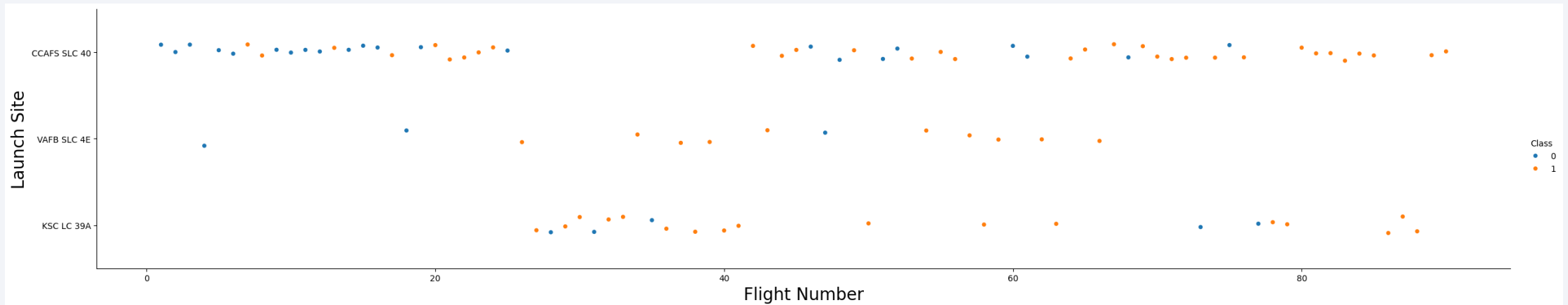
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



Section 2

Insights drawn from EDA

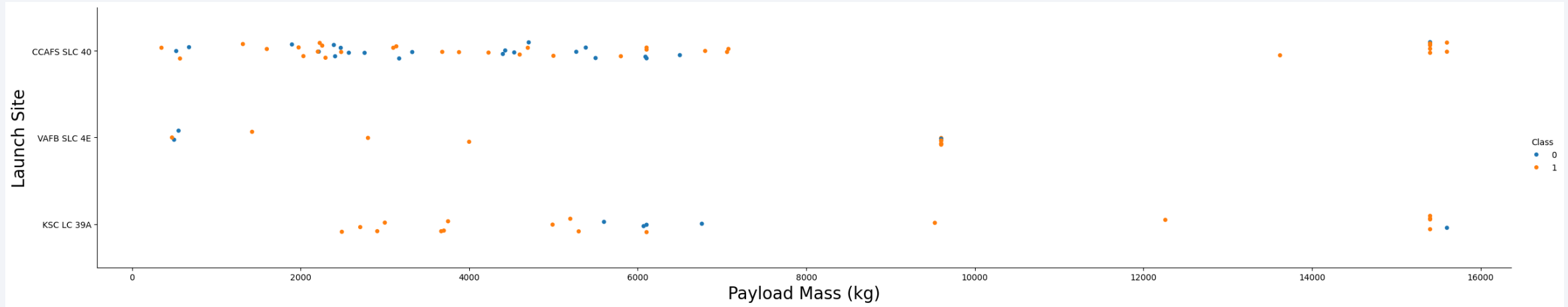
Flight Number vs. Launch Site



Explanation

- A large portion of the flights were launched from CCAFS SLC 40. The least frequent site is VAFB SLC 4E
- Apparently, flights from VAFB SLC 4E and KSC LC 39A have better results than CCAFS SLC 40.
- The previous statement may be biased as the CCAFS SLC 40 facility has more data points and the first flights were launched from there, so there are larger failure rates.
- After flight number 40 the failed/succeed ratio improves for the CCAFS facility.

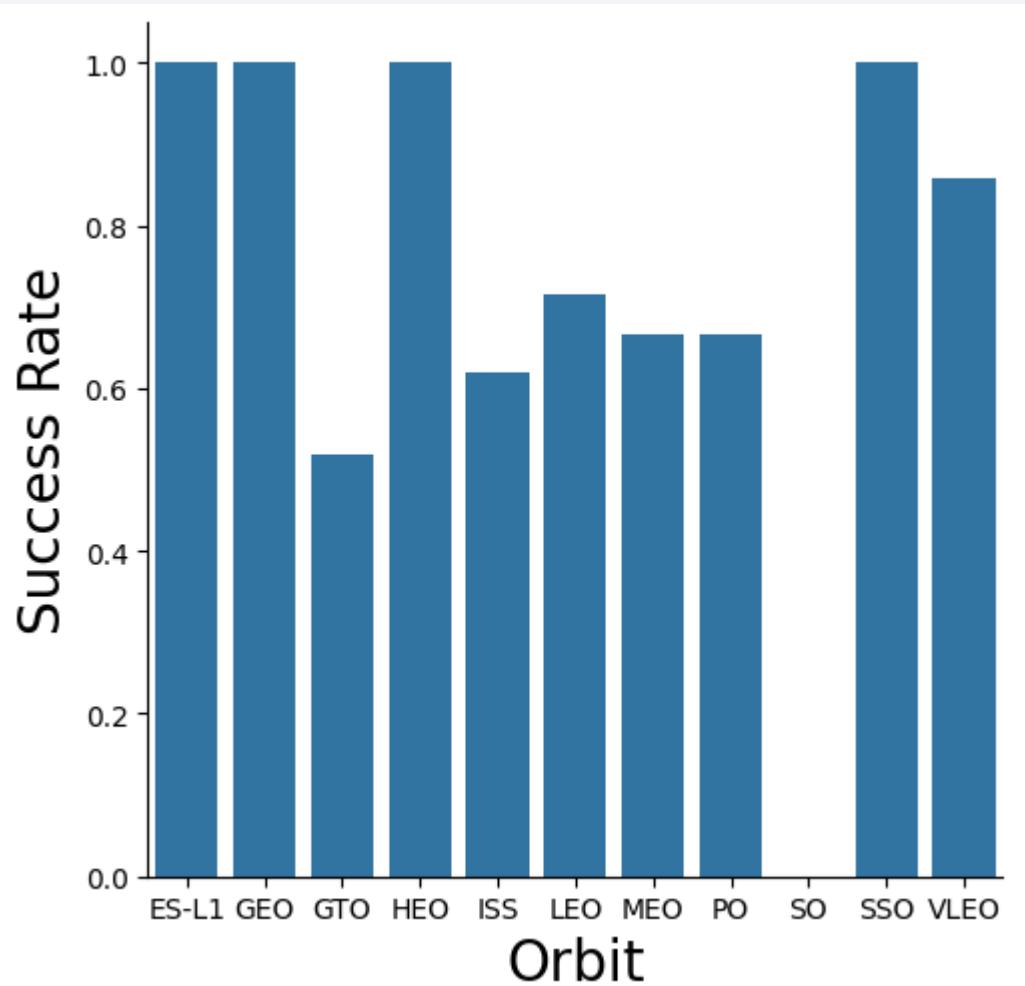
Payload vs. Launch Site



Now if you observe Payload Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

There isn't a clear correlation between these two variables

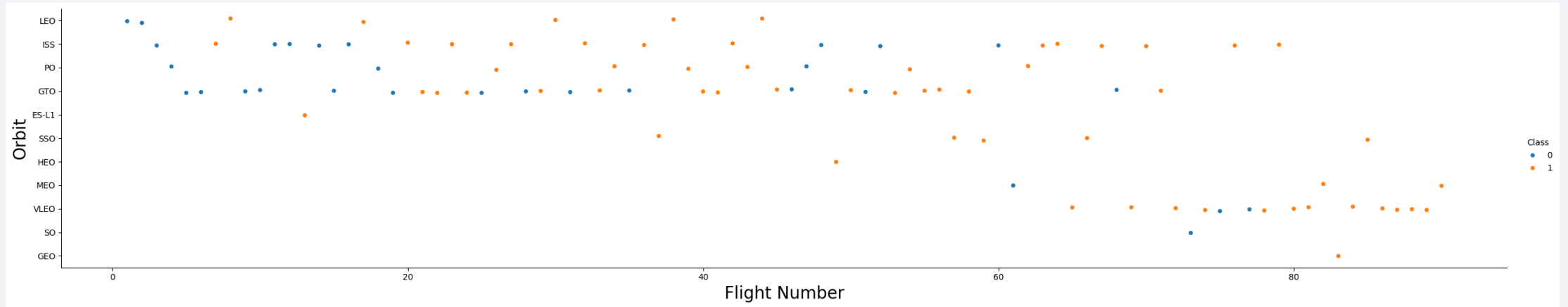
Success Rate vs. Orbit Type



Analyze the plotted bar chart try to find which orbits have high success rate.

- The 100% rate is achieved when orbits are ES-L1, GEO, HEO AND SSO.
- Orbits as ISS, LEO, MEO, PO and VLEO remain high.
- GTO Orbits are just about 50% successful.
- SO has 0% rate

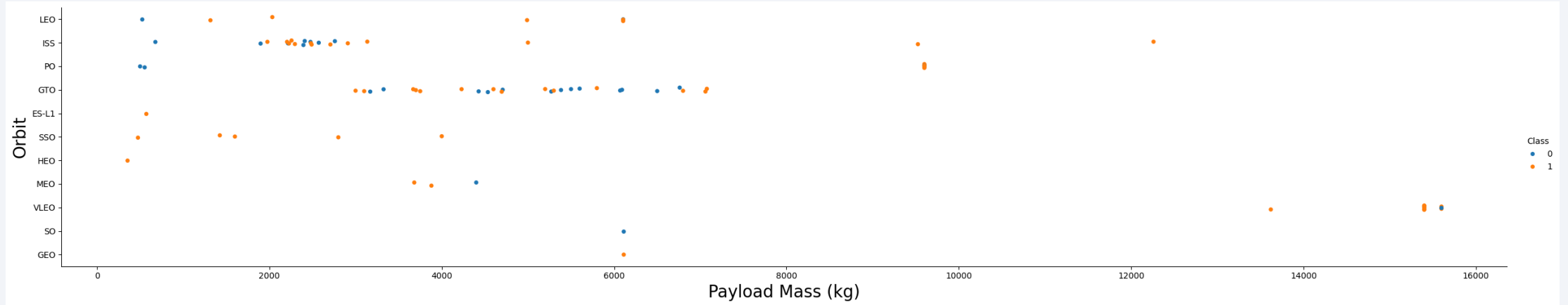
Flight Number vs. Orbit Type



You should see that in the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

GTO doesn't appear to be related with flight number.

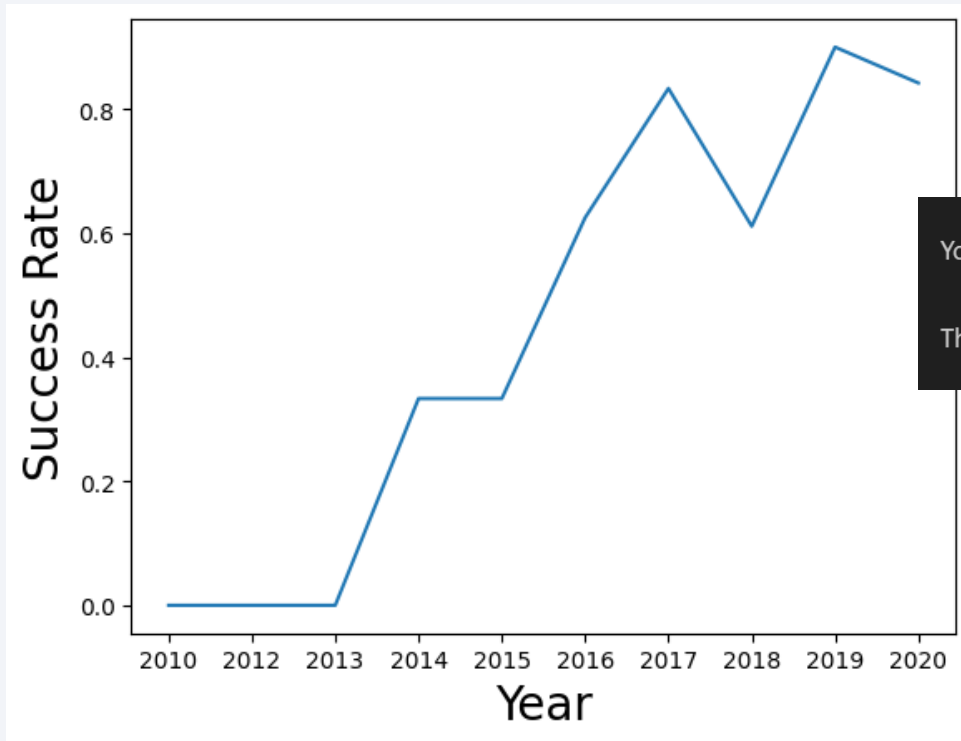
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.

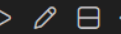
However for GTO we cannot distinguish this well as both positive landing rate and negative landing (unsuccessful mission) are both there here.

Launch Success Yearly Trend



You can observe that the success rate since 2013 kept increasing till 2017 (stable in 2014) and after 2015 it started increasing.

There was a low point in 2018, and also it lowers in 2020. May be related to the pandemic



Activar Windows

All Launch Site Names

- This query gives only unique values in specified column.

```
Display the names of the unique launch sites in the space mission
Generate + Code + Markdown

%sql select distinct launch_site from SPACEXTABLE

* sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Using 'LIKE' and the specified formatting returns values that begin with keyword 'CCA'.
- Limit 5 gives only 5 records

```
%sql SELECT * from SPACEXTABLE WHERE launch_site LIKE 'CCA%' limit 5;
```

Python

* [sqlite:///my_data1.db](#)

one.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- The SUM function performs a regular sum over the column.

```
%sql SELECT SUM(payload_mass__kg_) as total_payload_mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
total_payload_mass
```

```
45596
```

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- The AVG function performs the mean, using LIKE gets only the required booster version.

```
%sql SELECT AVG(payload_mass__kg_) as average_payload_mass FROM SPACEXTABLE where Booster_Version LIKE '%F9 v1.1%'
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
average_payload_mass
```

```
2534.6666666666665
```

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Using MIN() on a date field will return the first available date that complies with the conditions

```
%sql SELECT min(date) as first_successful_landing from SPACEXTABLE where landing_outcome = 'Success (ground pad)';
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
first_successful_landing
```

```
2015-12-22
```


Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- This query slices the data to get the values greater than 4000 and less than 6000 while getting the booster. The AND inside of the WHERE statement ensures this

```
%sql SELECT booster_version from SPACEXTABLE where landing_outcome = 'Success (drone ship)' and payload_mass_kg_ between 4000 and 6000

* sqlite:///my\_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- GROUP BY makes sure we get each outcome in a sole group. There's an outlier causing Success to be separated in two.

```
%sql SELECT mission_outcome, count(*) as total_number_success from SPACEXTABLE group by mission_outcome
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Mission_Outcome	total_number_success
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- MAX inside of the subquery gets the maximum payload. Then, the main query returns booster versions that carried said payload mass.

```
%sql SELECT booster_version from SPACEXTABLE where payload_mass_kg_ = (select max(payload_mass_kg_) from SPACEXTABLE)

* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Extracts outcome, booster version and launch site when the outcome is Failed in drone ship and year = 2015.

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

Generate

+ Code

+ Markdown

```
%sql SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5) = '2015' AND Landing_Outcome = 'Failure (drone ship)'
```

Python

```
* sqlite:///my_data1.db
```

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Groups the data, slices it between required dates, and counts the outcomes. Shows in descending order

```
Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
```

```
%sql SELECT Landing_Outcome, COUNT(*) as count_outcomes FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY count_outcomes DESC;
```

```
* sqlite:///my_data1.db  
Done.
```

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

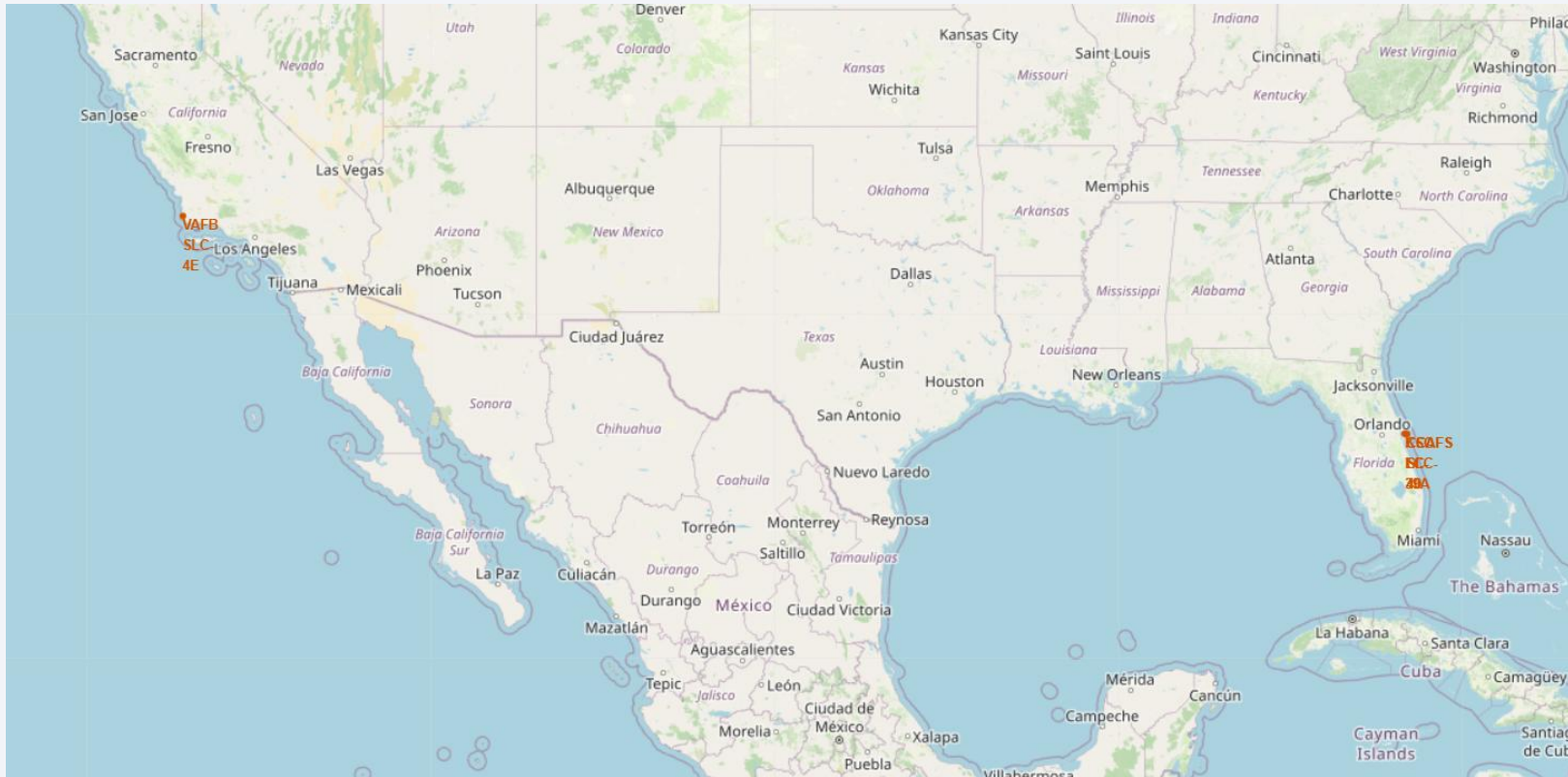
Reference Links

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue rectangle on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible, separating the dark surface from the deep blue of the atmosphere and the blackness of space.

Section 3

Launch Sites Proximities Analysis

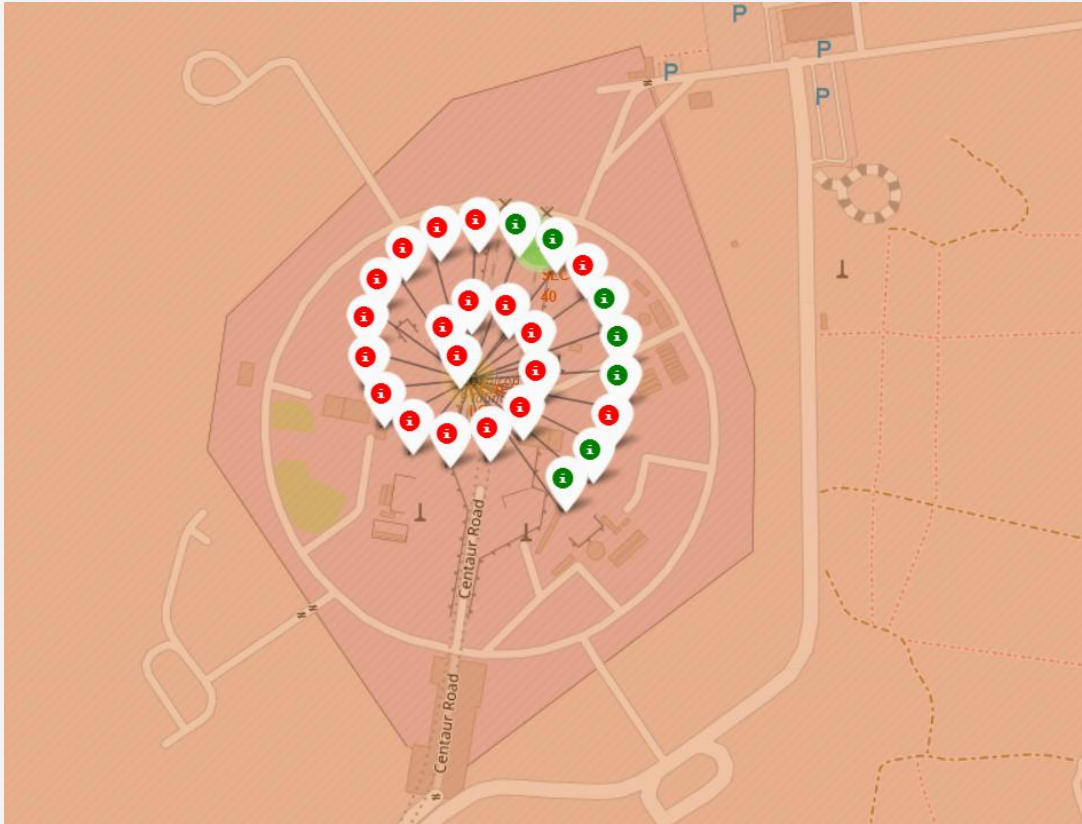
SpaceX Falcon9 – Launch Sites Map



Located launch sites:
California and Florida.
There are specifically
four locations:

- VAFB SLC-4E (CA)
- CCAFS LC-40 (FL)
- CCAFS SLC—40 (FL)
- KSC LC-39A (FL)

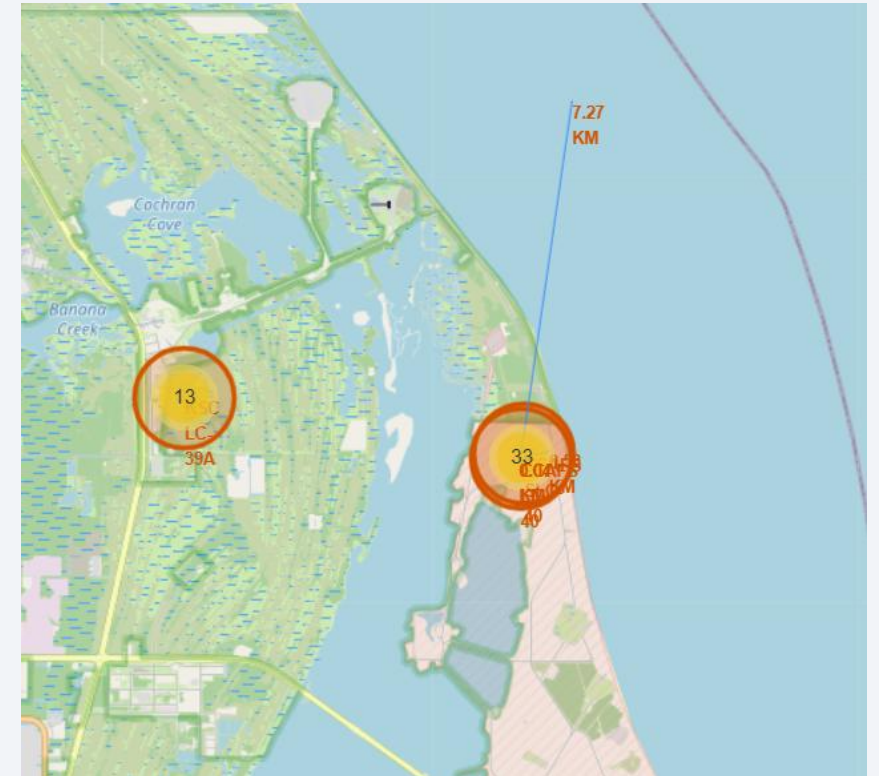
Success/Fail SpaceX Falcon9

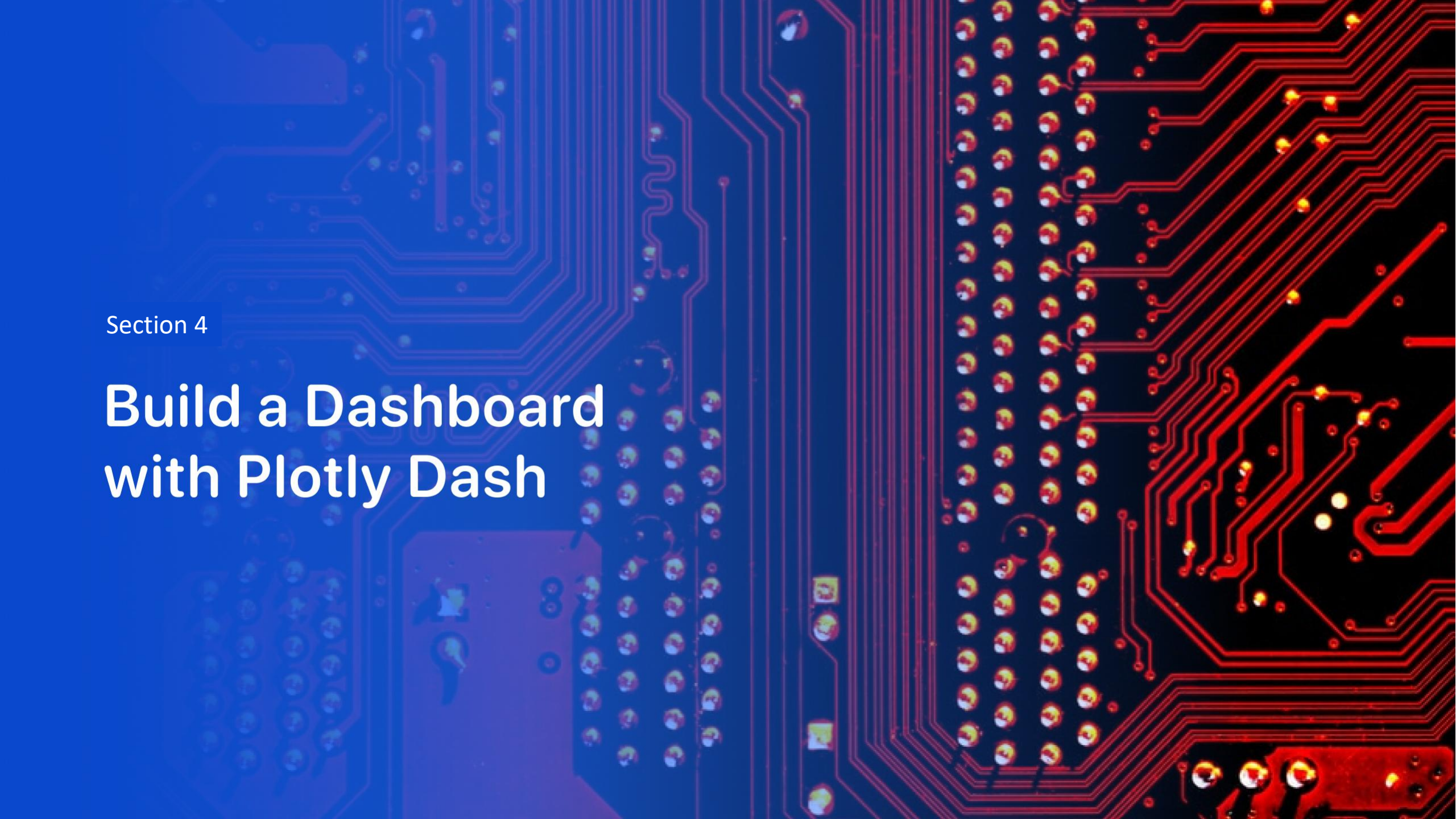


- This zoom on a location shows the markers added to indicate whether launch was successful (green) or failed (red).
- KSC LC-39A shows better markers for success
- CCAFS SLC-40 shows a greater number of launches.

SpaceX Falcon9 Proximities.

- The figure shows how close CCAFS LC-40 is to a coastline (7.27 km)





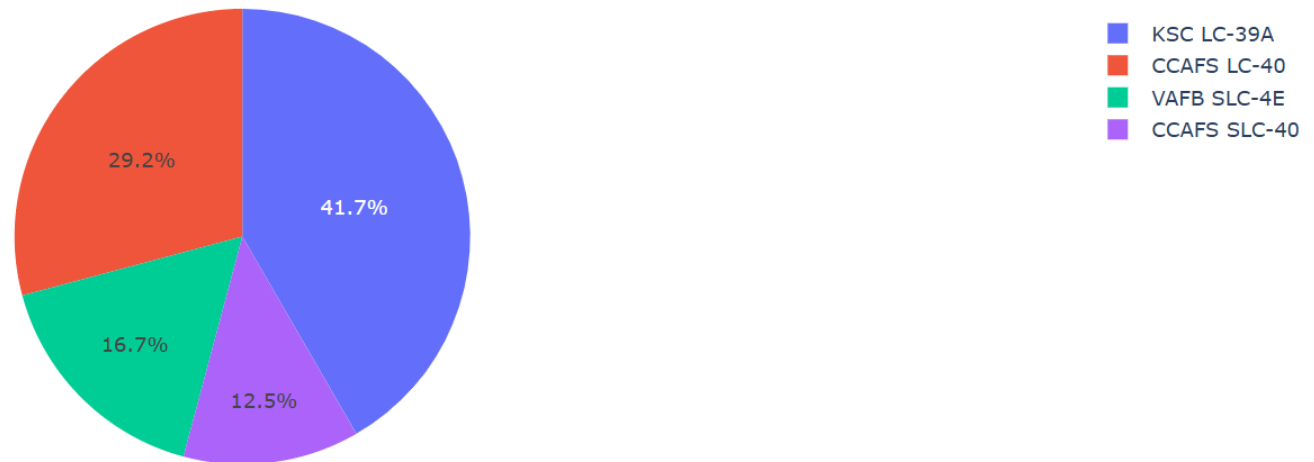
Section 4

Build a Dashboard with Plotly Dash

Plotly Dashboard – Launch Success

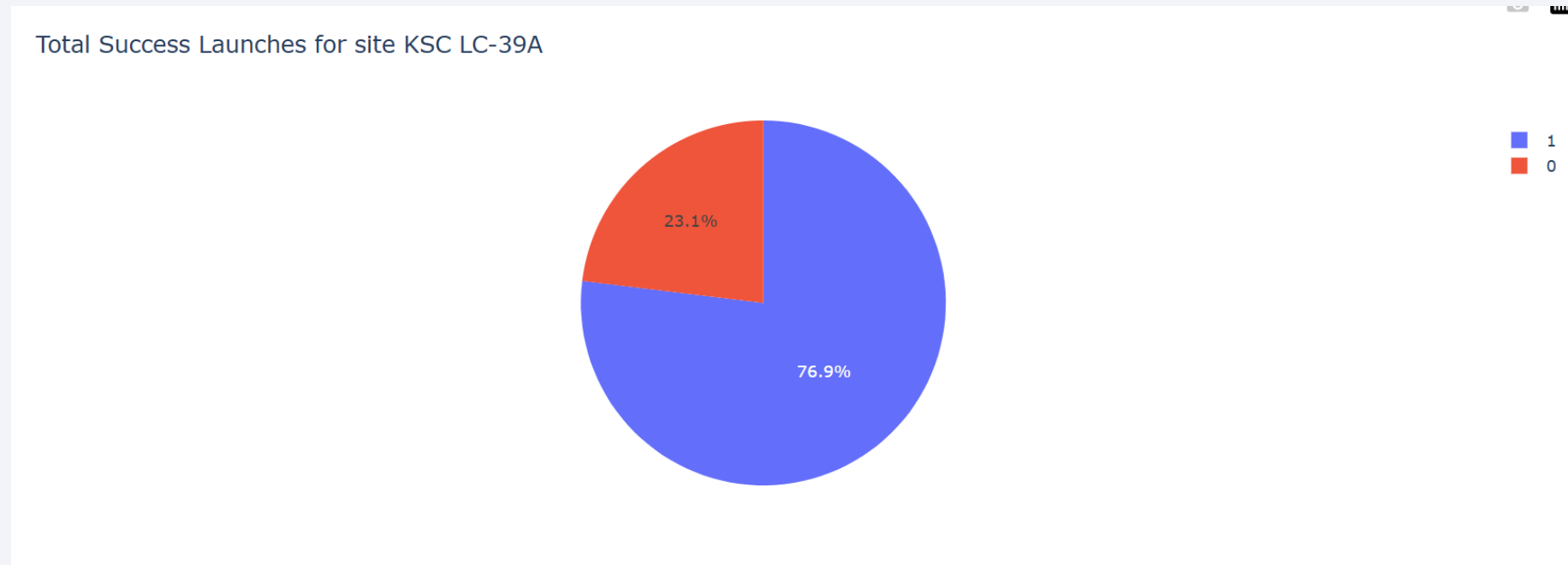
- It is clear that with 41.7% KSC LC-39A has the highest success rates, followed by CCAPS LC-40.
- CCAPS SLC-40 has the lowest success rates.

Total Success Launches By Site



<Dashboard Screenshot 2>

- When specifically reviewing KSC LC-39A the success rate goes to 76.9% against 23.1% failure rate



Plotly Dash – Payload Mass vs Success

- When Payload is over 5500 success tends to drop on all booster versions.

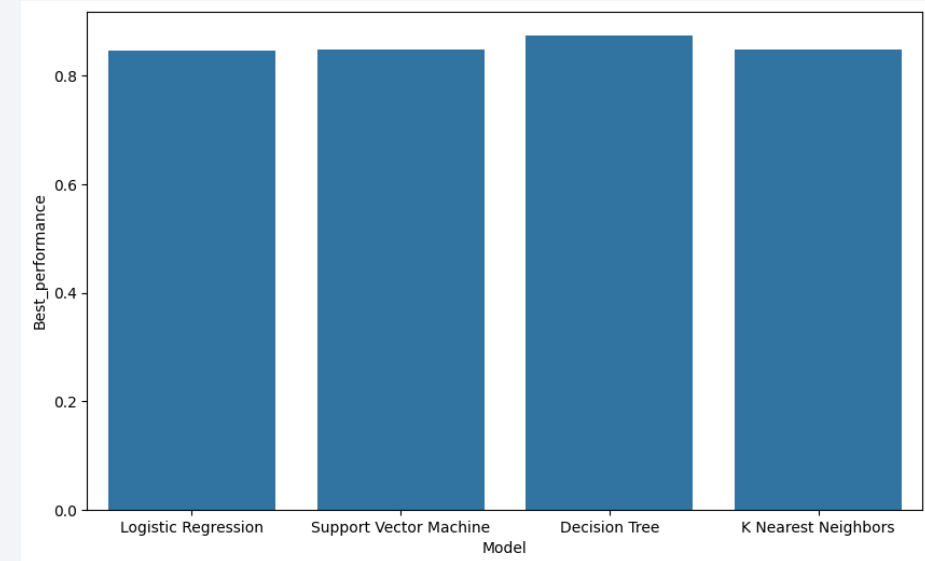


Section 5

Predictive Analysis (Classification)

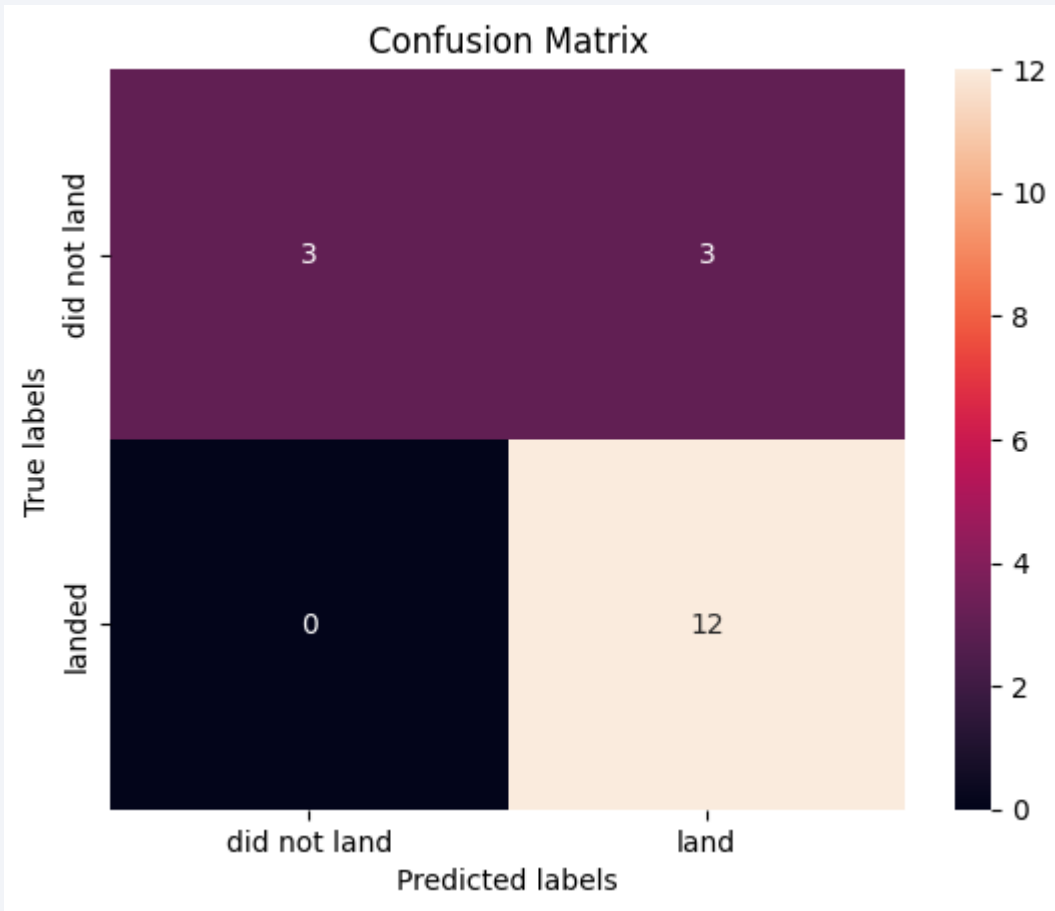
Classification Accuracy

- Even though, overall the models performed equally against the test dataset, the best performance obtained in the cross-validation was that of the Decision Tree, with 0.8767.
- Higher metrics could be achieved with a bigger dataset as the validation set was only 18 observations.
- A broader tuning grid could also help



Model	# Best_performance	# Test_Accuracy
Decision Tree	0.8767857142857143	0.8333333333333334
K Nearest Neighbors	0.8482142857142858	0.8333333333333334
Support Vector Machine	0.8482142857142856	0.8333333333333334
Logistic Regression	0.8464285714285713	0.8333333333333334

Confusion Matrix



- All models have the same confusion matrix, this is consistent with them having the same test accuracy. The models commit false positives, claiming a launch has successfully landed when it didn't'.

Conclusions

- As the bigger number of flights, the likelihood of a successful first stage increases
- Success rates and payload mass are not explicitly correlated, but a mild direct correlation might be present.
- KSC LC-39A is the top launch site, whereas CCAFS SLC-40 is underperforming.
- Launch sites are considerably far from cities, coastlines and railways, which helps assess risks and liabilities for possible failures.
- ML models can help predict the outcome of a launch, achieving up to 87% accuracy approximately.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

